

SULIT



UNIVERSITI TEKNOLOGI MALAYSIA
TEST 2 EXAMINATION SEMESTER I 2022 / 2023

SUBJECT CODE : SECR2043

SUBJECT NAME : OPERATING SYSTEM

SECTION : SECR/H/J/B/V/P / SCSR/J/B/V/P

TIME : 8.00 pm to 10.00 pm

DATE/DAY : 16 MAY 2023 (Tuesday)

VENUES : N24, DEWAN KULIAH 8 AND DEWAN KULIAH 9

INSTRUCTIONS:

There are **TWO parts**. Part A consists of 10 multiple-choice questions, and Part B consists of 3 structured questions related to computation and critical thinking. This test will contribute **25%** towards the total marks of 100%.

Warning!

Students who are caught cheating during the examination will be reported to the disciplinary board for possible suspension of the student for one or two semesters.

(Please Write Your Lecturer Name and Section)

Name	
I/C No.	
Year / Course	
Section / Lecturer Name (Circle)	[01] - Dr Farkhana [02] - Dr Syed Hamid Hussain Madni [03] - Dr Syed Hamid Hussain Madni [04] - Dr Zuriahati [05] - Dr Farkhana [07] - Dr Zuriahati

This question booklet consists of 8 pages, including the front page.

PART A: OBJECTIVES [10 MARKS]

Answer ALL questions.

1. Which of the following is/are shared by all the threads in a process?

I. Program counter

II. Stack

III. Address space ✓

IV. Registers

A. I and III

B. IV only

C. III and IV

☒ D. III only

2. Which of the following statement is TRUE for thread sharing?

☒ A. Both heap and global variables.

B. Global variable but not a heap.

C. Heap but not global variables.

D. Neither global variables nor heap.

3. *“It takes time and requires much space; allocating memory and resources for process creation is expensive.”*

The statement refers to one of the benefits of multithreading.

A. Responsiveness

☒ B. Economy

C. Scalability

D. Resource sharing

4. *“Programmers must ensure that the jobs provide equal work of equal value while finding activities that can execute in parallel.”*

The statement refers to one of the programming challenges.

☒ A. Balance

B. Data dependency

C. Identifying tasks

D. Data splitting

5. Which of the following critical section problem's solution avoids busy waiting while preserving mutual exclusion, progress and bounded waiting?

- A. Peterson algorithm ✓
- B. Test and Set ✗ bounded waiting
- C. Semaphore
- D. Swap

6. Based on the following code structure, what happens in the acquire lock section?

```
do {  
    acquire lock  
    critical section  
    release lock  
    remainder section  
} while (TRUE);
```

Enter CS / block in while

- A. An interrupt is disabled.
- B. The process is executed.
- C. The process gave a turn of execution to the following process.
- D. An interrupt is enabled.

7. Choose a TRUE statement of race condition.

- A. The problem which caused by the cooperating process.
- B. The problem which caused by the independent process.
- C. The problem which caused when two processes are executed in an atomic manner. ✗
- D. The problem which caused when two processes are executed using Semaphore. ✗

8. In a Multilevel queue, what scheduling algorithm can schedule the background processes?

- A. First Come, First Serve
- B. Round Robin
- C. Shortest Job Next
- D. Short Remaining Time

9. Select a scheduling algorithm that can be categorised into preemptive scheduling.

A. Shortest Remaining Time First

SRTF/SRT

B. First Come, First Serve

C. Shortest Job First

D. Shortest Job Next

I/O

10. The time taken to wait for the printer after the CPU execution is known as:

A. CPU burst *executing inst on the CPU*

B. I/O Burst

C. CPU scheduling *process deciding which process get the CPU next*

D. I/O request
asking for I/O service

PART B: STRUCTURED QUESTION [55 MARKS]

Answer all questions using the answer booklet provided.

QUESTION 1 [19 MARKS]

The following tables show the address for Processes (A, B and C) and Dispatcher in a memory. A single processor is used to run all these processes. Process A requests for I/O at the 4th instruction cycle and wait until the 25th instruction cycle. Next, at the 34th instruction cycle, Process B requests for I/O and remains for 22 instruction cycles.

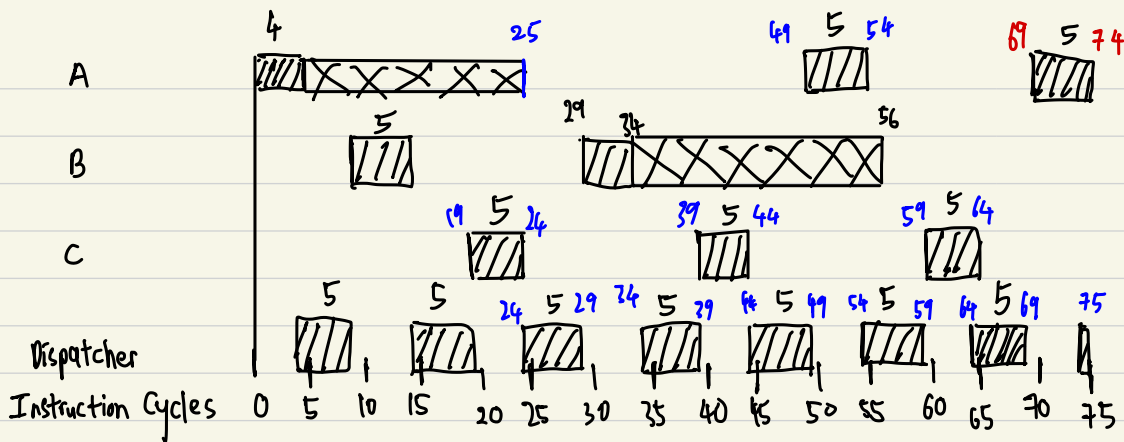
The Processes Location in the Memory

A	3000-3050	51
B	7000-7020	21
C	2000-2014	15
Dispatcher	200-204	5

All processes run up to 75 instruction cycles. Assume that the maximum CPU burst for each process is 5 instruction cycles. Calculate the following questions:

- Draw the Gant-Chart? [7]
- Which Job has been Unsuccessfully Executed? [2]
- What is the Waiting Time for Process C? [2]
- What is the Turnaround Time of Process C? [2]
- What are the States of processes A, B, and C at t_{42} ? [2]
- What is the Response Time of Process B? [2]
- What is the Throughput Time of Process C? [2]

(a)



(b) A and B

(c) Waiting time (PC) = ready

$$= 19 + (39 - 24) + (59 - 44)$$

$$= 19 + 15 + 15$$

$$= 49$$

$$@ 64 - 15$$

$$= 49s$$

(d) Turnaround time = 74 - 0

$$= 74$$

(e) t_{k2}

$P_c = \text{running}$

$P_A = \text{ready}$

$P_B = \text{blocked}$

(f) response time = 9s

(g) Throughput = 1

QUESTION 2 [21 MARKS]

- a) A fixed size **buffer of 7** fulfils the request between a system's **(Compiler and Assembler.)**
At $T = 0$, both processes share a standard variable named **"Counter"**, having an initial value = **5**. Complete **Table 1 (i - viii)** by updating the value of Counter and Event. [8]

Table 1

Time	Compiler	Counter	Assembler	Event
T0		5		Counter is initialized at 5
T1	1	<u>(i)</u> 6		Counter is <u>(v)</u> incremented by 1
T2		<u>(ii)</u> 5	1	Counter is <u>(vi)</u> decremented by 1
T3	1	<u>(iii)</u> 6		Counter is <u>(vii)</u> incremented by 1
T4	1	<u>(iv)</u> 7		Counter is <u>(viii)</u> incremented by 1

- b) Two processes are concurrently accessing their critical sections.
i. Using **Peterson's Algorithm**, complete **Table 2 (a - n)**. [7]

Table 2

Time	flag[0]	flag[1]	turn	Events
t_0	False	True	<u>(a)</u> 0	P1 request to enter CS1 <u>(b)</u>
t_1	False	<u>(c)</u> True	0	P1 enters CS1
t_2	True	True	1	P0 request to enter CS0 <u>(d)</u>
t_3	True	False	<u>(e)</u> 1	P1 executes RS1 <u>(f)</u>
t_4	True	False	<u>(g)</u> 1	P0 enters CS0 <u>(h)</u>
t_5	True	True	<u>(i)</u> 0	P1 request to enter CS1 <u>(j)</u>
t_6	False	True	<u>(k)</u> 0	P0 executes RS0 <u>(l)</u>
t_7	True	False	<u>(m)</u> 1	P0 request to enter CS0 <u>(n)</u>

- ii. From Question (i), when is the **bounded waiting** preserved? [1]

t_2 and t_5

- c) Based on the following statements, write out the possible value of semaphores and outputs for **Table 3**. Assume the following order sequence is executed between Process A and B under the pre-emptive scheduler. Any process continues until one of the semaphore values turns into 0. The initial value of semaphores for Process A (S) and Process B (Q) is set to be 3 and 2, respectively. [5]

3

2

//Process A While (1) { wait (S); printf ("Test 1 \n"); signal (S)	//Process B While (1) { wait (Q); printf ("SECR2043 \n"); signal (Q)
---	---

Table 3

(copy Table 3 to the answer booklet provided)

Process	S	Q	Output
Process B	2	3	SECR2043
Process A	2	3	Test 1
Process B	1	4	SECR2043
Process A	1	4	Test 1
Process B	0	5	SECR2043

QUESTION 3 [15 MARKS]

Table 4 shows six processes A to F arrived at different times with the length of CPU burst given in milliseconds (*ms*). Draw the **Gant-Chart** and calculate the **Average Waiting Time** and **Average Turnaround Time**.

Table 4

Processes	Arrival Time	CPU Burst	Priority
A	t_0	10	3
B	t_1	8	5
C	t_2	4	1
D	t_3	3	2
E	t_4	2	2
F	t_5	5	4

FCFS

- a) Assume the **FCFS Algorithm** is applied? [3] *Non*
- b) Assume the **SRT Algorithm** is applied? [3] *Pre*
- c) Assume the **SJN Algorithm** is applied? [3] *Non*
- d) Assume the **RR Algorithm** is applied with $q = 3$? [3] *Pre*
- e) Assume the **Priority Algorithm** is applied for all Processes A to F and arrives at time = t_0 with priority level one the highest priority? [3]

(d) FCFS

A	B	C	D	E	F
0	10	18	22	25	27
					32

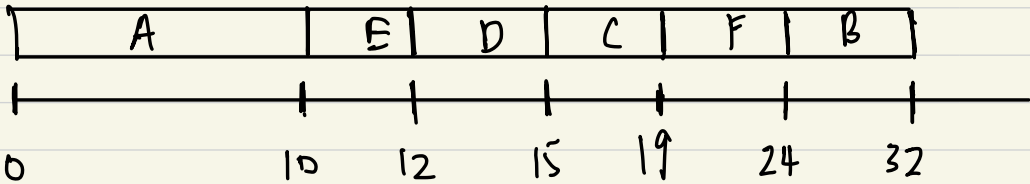
	Arrival Time	CPU Burst	Completion Time	Turnaround	Waiting
A	0	10	10	$10-0 = 10$	0
B	1	8	18	17	9
C	2	4	22	20	16
D	3	3	25	22	19
E	4	2	27	23	21
F	5	5	32	27	22
				<u>19.83</u>	<u>14.5</u>

(b) SRT

A	B	C	D	E	C	F	B	A			
0	1	2	3	4	5	6	8	11	16	23	32
A ₁₀	A ₉	A ₉	A ₉	A ₉	A ₉	A ₉	A ₉	A ₉	A ₉	A ₉	
	B ₈	B ₇	B ₇	B ₇	B ₇	B ₇	B ₇	B ₇	B ₇		
		C ₄	C ₃	C ₃	C ₃	C ₃	C ₃	F ₅			
			D ₃	D ₂	D ₁	E ₂	F ₅				
				E ₂	E ₂						
					F ₅						

	Arrival Time	CPU Burst	Completion Time	Turnaround	Waiting
A	0	10	32	32	22
B	1	8	23	22	14
C	2	4	11	9	5
D	3	3	6	3	0
E	4	2	8	4	2
F	5	5	16	9	4
				<u>13.17</u>	<u>7.83</u>

(C) SJN



B₈

~~C₄~~

~~D₃~~

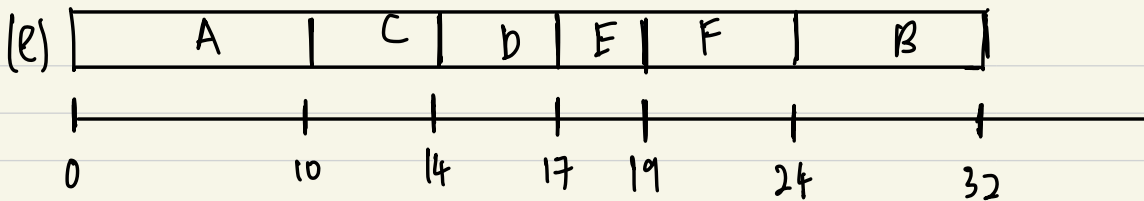
~~E₂~~

F₅

	Arrival Time	CPU Burst	Completion Time	Turnaround	Waiting
A	0	10	10	10	0
B	1	8	32	31	23
C	2	4	19	17	13
D	3	3	15	12	9
E	4	2	12	8	6
F	5	5	24	19	14
				<u>16.17</u>	<u>10.83</u>

[illegible]

	Arrival Time	CPU Burst	Completion Time	Turnaround	Waiting
A	0	10	30	30	20
B	1	8	32	31	23
C	2	4	24	22	18
D	3	3	12	9	6
E	4	2	14	10	8
F	5	5	26	21	16
				<u>20.5</u>	<u>15.17</u>



	Arrival Time	CPU Burst	Completion Time	Turnaround	Waiting
A	0	10	10	10	0
B	1	8	32	31	23
C	2	4	14	12	8
D	3	3	17	14	11
E	4	2	19	15	13
F	5	5	24	19	14
				<u>16.83</u>	<u>11.5</u>